Who Funds Open Data Sharing? Analysis of data availability statements in biomedical publications

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Abstract

Open data sharing is increasingly mandated by research funders, journals, and institutions, yet large-scale compliance measurement remains challenging. We analyzed $784{,}262$ open access biomedical research articles published between January 2024 and June 2025 using a dual-source pipeline: PDF-based text extraction (MinerU) where PDFs were available and PMC XML otherwise, followed by algorithmic detection of data sharing statements (oddpub v7.2.3), enriched with funder, journal, and institutional metadata from OpenAlex. The corpus included $256{,}660$ PDF-covered articles (32.7%) and $527{,}602$ XML-only articles (67.3%). We found an overall open data rate of 8.7%, rising to 16.6% among funded articles. Rates varied more than tenfold across the research ecosystem: major funders reached observed open data rates above 20%, while top journals reached observed rates of 70–86%, with corrected estimates as high as 92.9% (Nature Genetics) after adjusting for XML-only coverage limitations. PDF-based detection identified approximately 52% more data sharing statements than XML-based methods on the same articles. These observed rates differ markedly across funders and journals, and current overall sharing remains far below universal compliance. These patterns provide an empirical baseline against which future policy changes can be measured. An interactive dashboard at `https://www.opensciencemetrics.org` enables stakeholders to explore and benchmark these results.

1 Introduction

The open science movement has placed data sharing at the center of efforts to improve the transparency and reproducibility of research. The FAIR principles—that research data should be Findable, Accessible, Interoperable, and Reusable—have become a widely endorsed framework for scientific data stewardship (*1*). Meanwhile, surveys consistently reveal that a majority of researchers have encountered difficulties reproducing published results, underscoring the practical importance of access to underlying data (*2*). Studies have also demonstrated that publications linking to shared datasets receive more citations, creating incentive structures that complement institutional mandates (*3*).

In response, data sharing has become a shared priority of biomedical research funders worldwide, anchored by policies from the National Institutes of Health (*4*), the European Commission (*5*), UK Research and Innovation (*6*), the Wellcome Trust (*7*), the Deutsche Forschungsgemeinschaft (*8*), the Swiss National Science Foundation (9), the Agence Nationale de la Recherche (10), and the US National Science Foundation (11), as well as multilateral frameworks including the FAIR Data Principles (1) and the UNESCO Recommendation on Open Science (12). Funders and journals differ substantially in the stringency, scope, and enforcement of their data sharing policies, yet population-scale measurements of the open data rates that co-occur with these differences remain limited.

The scope and pace of these policies are generating new measurement demands. NIH's $1M Data Sharing Index ("S-index") Challenge, whose Phase 1 finalists were named in September 2025 (13, 14), seeks a researcher-level metric analogous to the h-index for data sharing; operationalizing such a metric at population scale will require exactly the kind of corpus-wide measurement infrastructure we describe here.

Journals have also emerged as influential gatekeepers of data sharing. Many high-impact journals now require data availability statements or deposit of data in public repositories as a condition of publication. Nature Research journals, Cell Press titles, and PLOS journals have progressively strengthened their data sharing policies over the past decade. However, the relationship between policy existence and actual compliance is not straightforward: previous work has shown that data availability statements do not always correspond to genuinely accessible data (15).

Measuring actual data sharing at scale is essential for tracking compliance with these policies and for benchmarking funders, journals, and disciplines against one another. Without systematic measurement, funders cannot benchmark their grantees' compliance, journals cannot assess the impact of editorial policies, and institutions lack the data needed to support their researchers. Yet scalable measurement presents significant methodological challenges. Manual assessment of data sharing is labor-intensive and impractical for large corpora (16). Text-mining approaches, while scalable, require access to the full text of the publication. The XML versions of articles in the PubMed Central Open Access (PMCOA) collection may miss data sharing statements present in final published versions data sharing info is often added late in the publication process or appears in sections omitted from the XML.

In this work we address these gaps by using a PDF-first detection pipeline. Full-text PDFs are processed using MinerU (17), a machine learning-based document extraction tool, and the resulting text is analyzed by oddpub v7.2.3 (18), an algorithm that identifies open data and open code sharing statements through pattern matching. This combination captures data sharing statements that XML-based methods miss, including those in supplementary sections, figure captions, and formatted data availability statements. On articles with both PDF and XML coverage, our PDF-based pipeline detects approximately 52% more open data statements than XML alone.

We analyze $784{,}262$ open access research articles from PubMed Central published between January 2024 and June 2025, characterizing data sharing rates across 816 funders and $1{,}389$ journals. Of these, $256{,}660$ articles had PDF-derived full text and $527{,}602$ were analyzed from XML only; journal-level correction factors from head-to-head PDF versus XML comparisons were applied to account for this differential coverage. We report observed open data rates (exact corpus proportions) alongside corrected estimates with 95% imputation intervals reflecting the uncertainty of the coverage correction. Our results reveal more than tenfold variation in data sharing rates across the research ecosystem, with clear signals that funder and journal policies are associated with substantially higher compliance.

All results are available through an interactive dashboard at https://www.opensciencemetrics.org, enabling funders, journal editors, institutional administrators, and researchers to explore the data across multiple dimensions.

2 Methods

2.1 Data Collection

We retrieved metadata for all open access research articles indexed in PubMed Central from January 2024 through June 2025, yielding a corpus of $784{,}262$ articles. Full-text PDFs were downloaded from the PMC Open Access subset where available ($256{,}660$ articles, 32.7%); the remaining $527{,}602$ articles (67.3%) were analyzed using XML full text. Article-level metadata—including journal of publication, funder information, and author institutional affiliations—was obtained from OpenAlex (*19*), an open bibliographic database that aggregates records from Crossref, PubMed, institutional repositories, and other sources.

2.2 Study Design

This is a cross-sectional, descriptive analysis. We compare observed open data rates across funders, journals, and disciplines within a single publication window using all open access PMC articles meeting our inclusion criteria. Because funders and journals are not randomly assigned to articles, and because funder, journal, and research discipline are tightly confounded in practice, the rate differences we report describe associations between entity-level context and open data rates; they do not identify the causal effects of any specific policy, mandate, or intervention. We make this scope choice deliberately: rigorous causal inference would require either a longitudinal pre/post design around a specific policy change or a quasi-experimental comparison with explicit controls, both of which we leave to future work. Statements throughout this paper about policy effectiveness, mandates "working," or community norms "driving" open data behavior should be read as descriptive co-occurrence, not causal estimation.

2.3 PDF Processing and Open Data Detection

PDFs were processed using MinerU (*17*), a machine learning-based extraction tool optimized for scientific documents that preserves document structure including tables, figures, and supplementary sections. The extracted text was then analyzed using oddpub v7.2.3 (*18*), an algorithm that detects open data and open code sharing statements in scientific publications through pattern matching against a curated dictionary of data sharing phrases, repository names, and accession number formats. Each article was classified as containing (or not containing) an open data statement and an open code statement.

For articles without available PDFs, we used the PMC XML full text as the input to oddpub. This dual-source approach maximized corpus coverage while enabling head-to-head comparison of detection rates between the two extraction methods.

2.4 Metadata Enrichment

Funder names reported in OpenAlex were normalized using a curated alias mapping covering 816 funders to aggregate related entities. For example, individual NIH institutes (e.g., NIMH, NIAID, NCI) were aggregated under their parent organization where appropriate, with child institute article counts deduplicated to avoid double-counting. This mapping was adapted from the funder normalization table used in the Open Science Metrics pipeline. OpenAlex works counts were used to filter for established funders, excluding small or niche entities with fewer than $100{,}000$ aggregated works from the main figures while retaining them in supplementary rankings.

Journal names were taken directly from the OpenAlex `primary_location.source.display_name` field.

2.5 Statistical Analysis

Open data rates were calculated as the proportion of articles with detected sharing statements, expressed as percentages. For the funder analysis, the baseline comparison rate was 16.6% (the open data rate across all articles with at least one identified funder). For the journal analysis, the baseline was 8.7% (the overall rate across all articles in the corpus).

2.5.1 Correction Factors for XML-Only Articles

Because PDF-based text extraction detects more data sharing statements than XML-based extraction, articles with only XML coverage have systematically underestimated open data rates. We developed a correction procedure using the subset of articles with both PDF and XML text (the "head-to-head" subset). For each journal with sufficient head-to-head data, we computed the "best" detection rate as the proportion of articles where *either* the PDF or XML extraction identified an open data statement. This best rate represents our estimate of the true open data rate for that journal, since it combines the complementary strengths of both extraction methods.

For articles with only XML coverage, we applied the journal-specific best detection rate as a correction factor. The corrected open data count for a given entity (funder or journal) was calculated as:

$$\hat{N}_{\text{OD}} = N_{\text{OD,PDF}} + \sum_{j} n_{j,\text{XML}} \times r_{j,\text{best}} \tag{1}$$

where $N_{\text{OD,PDF}}$ is the observed open data count among PDF-covered articles, $n_{j,\text{XML}}$ is the number of XML-only articles in journal j, and $r_{j,\text{best}}$ is the best detection rate for journal j from the head-to-head subset. For journals without sufficient head-to-head data, a global fallback rate was used. The corrected estimate was floored at the observed count to ensure that corrections never reduce an already-observed signal.

Because this analysis is a census of all articles meeting our inclusion criteria rather than a sample drawn from a larger population, observed open data rates are exact proportions for the corpus and are reported without confidence intervals. The only quantity carrying statistical uncertainty is the XML-only correction, which *imputes* the open data statements that PDF-based processing would have detected in articles for which only XML was available. We quantify this uncertainty with a 95% *imputation interval*: a Wilson score interval (*20*) on the head-to-head best detection rate, propagated through the weighted correction of Equation (1). This interval should be read as the uncertainty introduced by the coverage correction, not as sampling uncertainty of the headline rate. Where the correction does nothing—an entity with no XML-only articles, or one whose imputed estimate is floored at the observed count—no imputation interval is reported.

2.5.2 Figure Inclusion Thresholds

To focus figures on well-powered entities, we applied Weibull-derived survival thresholds to determine the minimum article count for inclusion. Funders required at least $2{,}586$ funded articles and

$100{,}000$ aggregated OpenAlex works for the main figure; journals required at least $1{,}832$ articles. These thresholds correspond to fixed survival quantiles (the article count at which the fitted Weibull survival function equals the stated percentage) of a Weibull distribution fitted to entity-level article counts, ensuring that displayed rates are based on sufficient sample sizes. All entities above the minimum analysis threshold (100 articles for funders) are included in the supplementary tables.

3 Results

3.1 Overall Open Data Rates

Across our corpus of $784{,}262$ open access biomedical research articles published from January 2024 through June 2025, we detected open data sharing statements in 8.7% of publications, establishing a baseline rate for recent biomedical literature. Among the subset of articles with at least one identified funder, the rate was 16.6%, nearly double the overall baseline, suggesting that funding is itself associated with higher data sharing.

3.2 Variation by Funder

Open data rates varied substantially across the 816 funders analyzed, and which funders lead depends on how the set is filtered. We therefore distinguish two strata. Among the *major world funders* that account for the bulk of global research output—those exceeding both the Weibull article-count threshold and $100{,}000$ aggregated OpenAlex works, shown in Figure 1—the leaders reached roughly 1.5 times the 16.6% funded-article baseline. Among *all* 816 funders meeting the minimum analysis threshold (Table **??**), the highest per-article rates were instead achieved by smaller, specialized research organizations, reaching more than three times the baseline. Funder size and per-article open data rate are thus largely decoupled—a point we return to in the Discussion—so we describe each stratum in turn.

Among the major world funders shown in Figure 1, the Agence Nationale de la Recherche (France) led at 26.1% observed, followed by the U.S. National Science Foundation (24.7%), the Wellcome Trust (24.1%), the Swiss National Science Foundation (23.4%), and the Deutsche Forschungsgemeinschaft (21.6%). The U.S. National Institutes of Health—by far the largest funder in the corpus ($28{,}982$ funded articles)—reached 20.5%, and UK Research and Innovation 19.4%. Every major funder in this stratum exceeded the 16.6% funded-article baseline, but none approached the rates of the smaller, specialized organizations described next: the largest funders are not the highest per-article performers.

Across the full set of 816 funders (Table **??**), the highest observed rates were concentrated among smaller, mission-focused research organizations. Swedish organizations led these rankings: the Uppsala Multidisciplinary Center for Advanced Computational Science (UPPMAX) achieved the highest observed rate at 52.7% (59 of 112 articles), followed by the Science for Life Laboratory (SciLifeLab) at 49.1% (108 of 220 articles). Both organizations are closely affiliated with Sweden's national bioinformatics infrastructure, where data sharing is deeply embedded in research workflows.

European research organizations were prominently represented among the top performers. The European Molecular Biology Laboratory (EMBL) achieved 46.7% (57 of 122 articles), the Francis Crick Institute 40.1% (73 of 182), and the European Molecular Biology Organization (EMBO) 36.7%

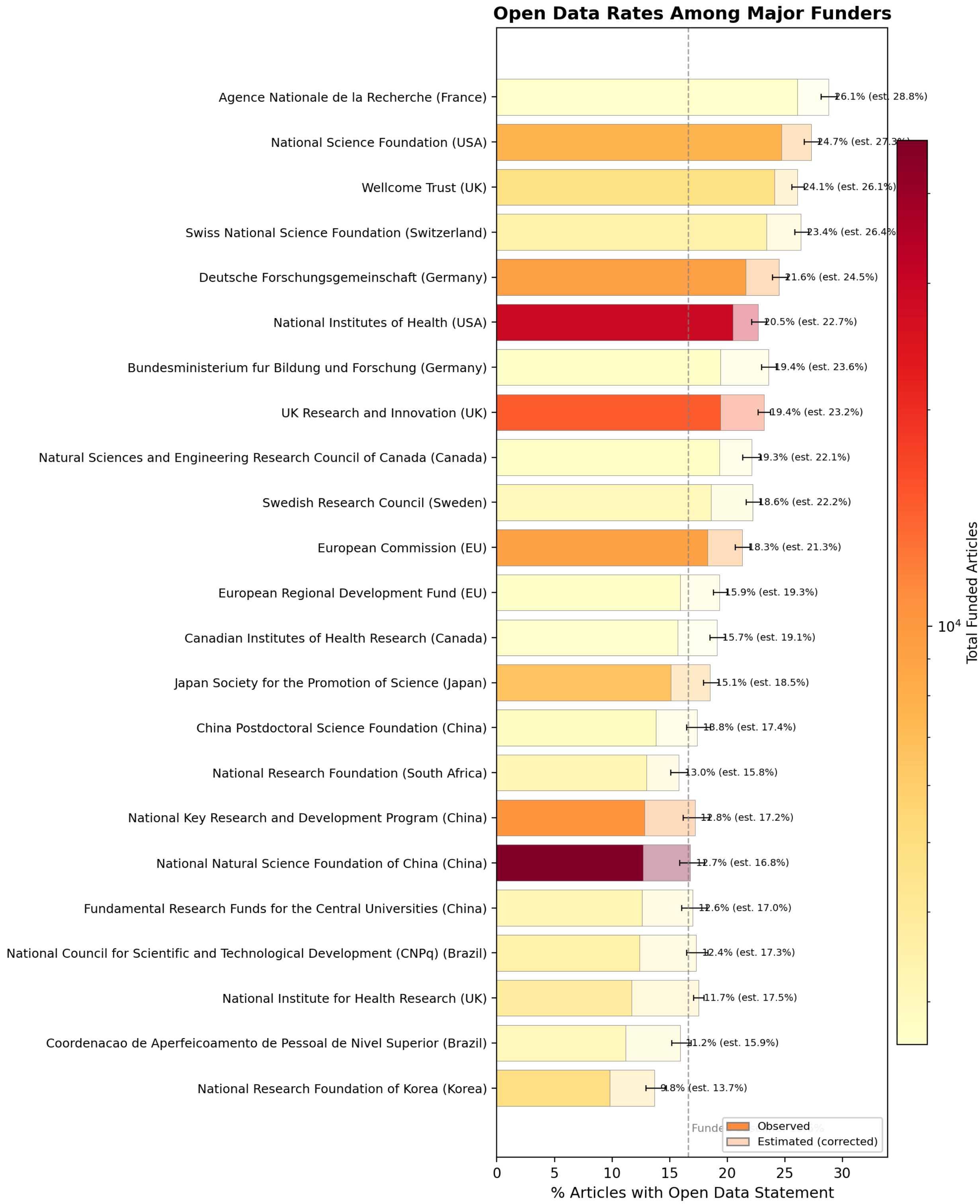

Figure 1: Open data rates among major biomedical research funders (2024–2025 research articles). Bar length shows the observed open data rate (full opacity) and the estimated corrected rate (lighter shade) after applying journal-level PDF vs. XML correction factors. Error bars indicate the 95% imputation interval on the corrected estimate (uncertainty in the XML-only coverage correction, not sampling uncertainty of the observed rate); entities where the correction does nothing carry no interval. Bar color encodes total funded article count on a log scale (yellow = fewer, red = more). Dashed line shows the funded-article baseline. Only funders exceeding both the Weibull-derived 3% survival article-count threshold and 100,000 aggregated OpenAlex works are shown.

(106 of 289). German funders including the Deutsches Krebsforschungszentrum (38.7%), the Leibniz Association (33.8%), and the Max Planck Society (30.6%, 293 of 958 articles) all substantially exceeded the baseline. French funders including the Fondation ARC pour la Recherche sur le Cancer (37.9%), the European Synchrotron Radiation Facility (37.8%), and INRAE (35.7%) showed a similar pattern.

Among United States funders, the Department of Energy's Biological and Environmental Research program achieved 40.6% (108 of 266 articles) and the Argonne National Laboratory reached 43.6% (68 of 156). The NIH Common Fund (35.6%), the NSF Directorate for Biological Sciences (34.8%, 288 of 828 articles), and the DOE Office of Science (31.3%, 233 of 744) all demonstrated rates well above the funded-article baseline. Private philanthropies also performed strongly: the Howard Hughes Medical Institute achieved 34.9% (261 of 748 articles) and the Burroughs Wellcome Fund 32.5%.

Canadian funders including the Canadian Institute for Advanced Research (42.9%) and Genome Canada (35.0%) appeared in the top 20, as did the Swiss École Polytechnique Fédérale de Lausanne (32.9%) and ETH Zürich (28.3%).

3.3 Variation by Journal

Journal-level open data rates showed even greater variation than funder rates, spanning from below 5% to above 90% estimated (Table 2, Figure 2).

Journals in genomics and structural biology dominated the top ranks. Nature Structural & Molecular Biology led with an observed rate of 86.1% (130 of 151 articles) and an estimated rate of 91.7% after correction for XML-only coverage. Microbial Genomics (84.8%), Molecular & Cellular Proteomics (81.6%), Cell Genomics (80.2%), and Genome Research (76.0%) all exceeded 75% observed. Nature Genetics achieved the highest estimated rate among all journals at 92.9% (95% imputation interval: 91.5–93.7%), reflecting the substantial uplift from PDF-based detection over its predominantly XML-only coverage. Genome Biology reached an estimated 89.3% (95% imputation interval: 88.1–90.2%).

A notable cluster of psychology and cognitive science journals also demonstrated high open data rates: the Journal of Cognition (72.8%), Attention, Perception, & Psychophysics (62.9% observed, 68.2% estimated), Psychological Research (62.2%), Behavior Research Methods (59.8%), and Communications Psychology (54.1%). These rates co-occur with the growing open data culture in experimental psychology, where community norms and journal requirements emerged in response to the replication crisis in that field.

Among the Nature portfolio, open data rates ranged from 86.1% (Nature Structural & Molecular Biology) to 45.2% (Nature Communications), reflecting the diverse disciplinary scope of these journals and varying levels of field-specific data sharing norms. Nature Immunology (62.8% observed, 79.1% estimated), Nature Cell Biology (62.1% observed, 76.3% estimated), Nature Methods (59.0% observed, 73.7% estimated), and Nature Neuroscience (55.5% observed, 69.8% estimated) all showed substantial upward corrections, indicating that PDF-based detection captured data sharing statements missed by XML extraction in these journals.

High-volume journals provide insight into data sharing at scale. Scientific Data published $2{,}421$ articles in our window with a 65.6% open data rate, consistent with its mission as a dedicated data journal. iScience ($3{,}514$ articles, 50.8%), Communications Biology ($2{,}503$ articles, 47.2% observed,

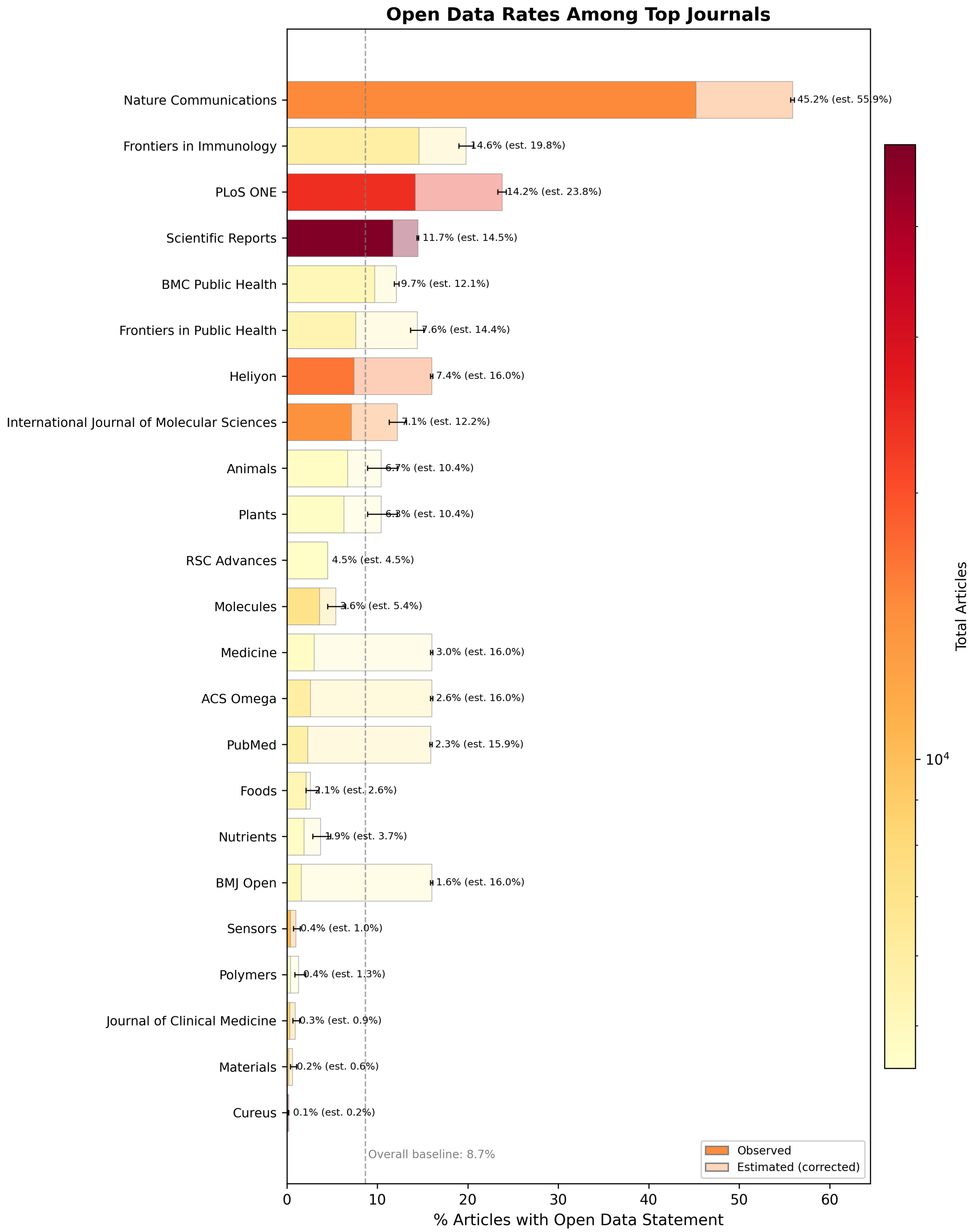

Figure 2: Open data rates among top biomedical journals (2024–2025 research articles). Solid bars show the observed open data rate; lighter background bars show the estimated (corrected) rate after applying journal-level correction factors from head-to-head PDF vs. XML comparison. Error whiskers indicate the 95% imputation interval on the corrected estimate (uncertainty in the XML-only coverage correction, not sampling uncertainty of the observed rate); journals where the correction does nothing carry no whisker. Bar color encodes total article count on a log scale (yellow = fewer, red = more). Dashed line shows the overall baseline. Only journals exceeding the Weibull-derived survival article-count threshold are shown.

60.1% estimated), and Nature Communications ($15{,}129$ articles, 45.2% observed, 55.9% estimated) demonstrate that moderately high rates are achievable even at very large publication volumes.

3.4 Interactive Data Exploration

> **Interactive Dashboard:** Readers can explore these results dynamically through our web dashboard at `https://www.opensciencemetrics.org`, which provides real-time filtering and visualization across multiple dimensions including time periods, countries, and research domains.

4 Discussion

4.1 Key Findings

Our analysis of $784{,}262$ open access biomedical articles reveals that open data sharing is not uniformly distributed across the research ecosystem. The overall rate of 8.7%—and even the 16.6% funded-article baseline—falls far short of universal compliance with data sharing mandates. Yet the more than tenfold variation we observe, from single-digit percentages to above 90%, shows that high data sharing rates do occur in some funder and journal contexts, even if our cross-sectional design cannot say what drives that variation.

High open data rates at both the funder and journal level co-occur with two factors visible in our data: research domains where data sharing is technically routine (genomics, structural biology, environmental science) and funder or journal contexts with explicit, enforced sharing requirements. Funders with strong institutional cultures of openness (e.g., EMBL, SciLifeLab, the Francis Crick Institute) reach rates two to three times the funded-article baseline. Among journals, the pattern is more pronounced: journals that require data deposit as a condition of publication (e.g., Nature Genetics, Genome Research) reach rates above 75%, while journals with weaker or absent data sharing requirements remain near or below the baseline. Because funders and journals are not randomly assigned to articles, and because policy and discipline are tightly confounded, we cannot attribute these differences to policy text as such.

4.2 Comparison to Previous Work

Our results extend and refine previous large-scale assessments of research transparency. Serghiou et al. documented the prevalence of multiple transparency indicators across biomedical publications using XML-based text mining (*21*). Our lower overall rate of 8.7% likely reflects both the larger and more heterogeneous corpus (including articles from a broader range of journals) and the specificity of the oddpub v7.2.3 detection algorithm. Importantly, our PDF-based detection pipeline identifies approximately 52% more open data statements than XML-based extraction on the same articles, demonstrating that previous XML-only estimates systematically undercount data sharing. This improvement is attributable to MinerU's ability to extract text from formatted data availability statements, supplementary sections, and other PDF elements not represented in PMC XML markup.

The correction factors we apply to account for XML-only coverage provide estimated rates that are consistently higher than observed rates, particularly for journals where a large fraction of articles lack PDF extraction. For example, Nature Genetics' observed rate of 70.5% rises to an estimated

92.9% after correction, reflecting the journal's strong data sharing culture that is partially masked by incomplete PDF coverage.

The PLOS Open Science Indicators dataset (v10; 138,995 PLOS articles and 27,868 comparator articles with publication dates through 31 March 2025) provides a contemporary external benchmark produced by PLOS in partnership with DataSeer from journal XML (22).[1] The continued use of XML-based extraction in widely cited monitoring datasets underscores the value of the PDF-first approach we adopt for PMC-scale measurement.

Open data detection is also beginning to shift toward large language model (LLM) approaches operating on full text. A March 2026 pilot reported that an LLM applied to a corpus of approximately 6,000 Michael J. Fox Foundation–funded articles detected evidence of data *reuse* in 43% of articles, compared with approximately 2% via traditional identifier-based methods (23). That comparison addresses a different task than ours (reuse rather than declared sharing), and the underlying model is closed-source, but it indicates the direction of methodological change. We view keyword- and pattern-based pipelines such as oddpub as the appropriate open-source, auditable, institution-scale baseline against which future LLM-based detectors should be benchmarked.

Changes from preliminary results: Some funders that featured prominently in our earlier presentation of these results no longer appear in the main figure. Most notably, the Howard Hughes Medical Institute (HHMI)—a private philanthropy that funds a relatively small number of elite investigators—achieved one of the highest per-article open data rates in the dataset (34.9%, ranked 18th of 816 funders). However, HHMI's 748 funded articles in the 2024–2025 window fall well below the Weibull-derived figure threshold of $\geq$2,586 articles, and its 40,261 total OpenAlex works fall below the 100,000-work minimum applied to exclude small or niche funders. These thresholds, while necessary for focusing the figure on major world funders, systematically exclude high-performing private foundations with narrow funding portfolios. HHMI and similar funders (e.g., the Simons Foundation, the Chan Zuckerberg Initiative) remain visible in the full supplementary rankings. This pattern underscores that funder *size* and funder *effectiveness* at promoting open data are distinct dimensions that should be interpreted separately.

4.3 Implications for Policy

These results have several implications for research policy, though they should be read as descriptive patterns rather than causal evidence of policy effects. First, funders and journals with explicit, enforced data sharing requirements consistently appear at the top of the rankings, an association compatible with mandates working but also with selection of compliance-friendly researchers into those funders and journals. Second, variation across funders with nominally similar mandates suggests that stringency and enforcement may matter more than the bare existence of a policy — a hypothesis that would need within-funder longitudinal data to test directly. Third, the disciplinary clustering we observe (genomics and psychology journals leading their respective tiers) indicates that community norms co-vary with measured open data rates beyond what any top-down policy text would predict.

Our results also provide a benchmarking framework. Funders can compare their grantees' data sharing rates against peer organizations, journals can assess their position relative to comparable

[1] The OSI release is cumulative from 2018 through March 2025 and covers PLOS journals plus matched non-PLOS comparators; its `Data_Shared` and related indicators are not directly comparable to oddpub-based rates on our PMC corpus.

publications, and institutions can identify areas for targeted investment in data management infrastructure and training. The interactive dashboard at `https://www.opensciencemetrics.org` is designed to support exactly these comparisons.

These findings are timely given parallel NIH efforts to build researcher-level incentives for data sharing. The Data Sharing Index Challenge, with Phase 2 deliverables due in mid-2026, will produce candidate metrics that score individual investigators on data sharing quality and reuse (*13*, *14*). Whatever metric emerges, validating it at population scale will require independent corpus-derived rates of the type our pipeline produces, suggesting a natural complementarity between top-down incentive design and bottom-up behavioral measurement. The implementation environment is also evolving: NIH released NOT-OD-26-046 in February 2026, introducing a structured (largely yes/no) Data Management and Sharing Plan format effective May 25, 2026 (*24*). While the underlying 2023 DMS Policy is unchanged and this format change affects pre-award plans rather than the published-article content our pipeline measures, it is likely to make pre-award compliance signals more amenable to machine extraction and easier to link to retrospective sharing rates of the kind reported here.

4.4 Limitations

Several limitations should be noted. First, our corpus is restricted to open access articles available through PubMed Central; closed-access articles may exhibit different data sharing patterns, likely biased toward lower rates. Second, external validation at the University of Edinburgh (*25*) found high precision for detecting data sharing statements, but sensitivity of only 61.8% for open data detection (and 30% for open code) when a data availability statement was present, materially below the 0.73 sensitivity reported in the original 2020 validation (*18*). Our reported rates therefore likely underestimate true declared sharing, even before considering non-standard phrasings or implicit data sharing (e.g., data available “upon request”). Third, detecting a data sharing *statement* is not equivalent to verifying that data are actually accessible and reusable; our rates should be interpreted as upper bounds on genuine data availability. A recent large-scale audit of four leading medical journals (BMJ, JAMA, NEJM, Lancet) found that 98.0% of papers do not actually share available research data despite policies requiring or mandating sharing (*26*), underscoring this statement-versus-practice gap. Fourth, the correction factors assume that the head-to-head subset (articles with both PDF and XML) is representative of XML-only articles within each journal, which may not hold uniformly. Fifth, our January 2024–June 2025 window captures a snapshot of current practices but does not permit longitudinal analysis of trends over time. Sixth, and most fundamentally, the analysis is cross-sectional with no counterfactual: funders and journals are not randomly assigned to articles, so any rate differences across them reflect a confounded mixture of policy text, enforcement, discipline, infrastructure, and researcher self-selection. Distinguishing the contribution of any one factor would require either a longitudinal pre/post design around a specific policy change or a quasi-experimental comparison with explicit controls.

4.5 Future Directions

Future work will increase PDF extraction coverage for the XML-only portion of the current $784{,}262$-article corpus and extend the temporal window beyond June 2025. Longitudinal analysis will enable assessment of whether specific policy changes (e.g., the 2023 NIH Data Management and Sharing Policy) are associated with measurable increases in data sharing rates. Institution-level and data-repository-level analyses are reserved for a follow-up paper, where the substantial methodological work of name normalization (for institutions) and repository-mention extraction (for data repositories) can be treated in the depth they require. A further open question is the head-to-head performance of contemporary LLM-based detectors against the oddpub regex/keyword pipeline on the binary "is open data" task; no such benchmark has yet been published, and the corpus and head-to-head subset described here are well suited to serve as the evaluation substrate.

4.6 Conclusion

This study demonstrates that open data sharing in biomedicine varies more than tenfold across funders, journals, and disciplines. Many major funders exceeded the 16.6% funded-article baseline, with top-ranked organizations reaching observed rates of 30–53%; leading journals reached estimated rates above 90%, compared to an overall baseline of 8.7%. These findings are consistent with policy mandates and community norms being associated with higher data sharing, while also highlighting the considerable gap between current practice and universal compliance. Continued investment in scalable measurement tools, such as the pipeline and dashboard presented here, will be essential for monitoring progress toward a more open and reproducible research enterprise.

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This work utilized the computational resources of the NIH HPC Biowulf cluster (`https://hpc.nih.gov`).

Data Availability

Full results including all funder and journal rankings are available as supplementary CSV files in the project repository. The interactive dashboard at `https://www.opensciencemetrics.org` provides dynamic exploration of all results. The oddpub detection pipeline and analysis code will be made available upon publication.

Use of Generative AI

The authors used generative AI tools during the preparation of this manuscript, including Anthropic's Claude (via claude.ai and Claude Code), OpenAI models, and the Cursor agentic code editor. These tools were used as drafting, editing, and coding assistants: initial prose for several sections was generated from the author's outlines, analysis notes, and pipeline outputs, then reviewed, revised, and edited by the author over the course of manuscript preparation. AI assistants were also used during development of the analysis pipeline and supporting code. The author verified all numerical results against the underlying analysis outputs, confirmed that all cited references exist and support the claims made, and is responsible for the accuracy and integrity of the final text. Generative AI was not used to fabricate, alter, or substitute for the underlying data. The full revision history is available in the project's public Git repository https://github.com/nimh-dsst/open-science-metrics.

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A Supplementary Tables

Table 1: Open data rates among major biomedical research funders. Funders exceeding the Weibull-derived 5% survival threshold for total funded articles (≥0 articles with oddpub v7 coverage), ranked by observed open data rate. Parent funders (e.g., NIH, UKRI) aggregate all child institutes with deduplicated article counts. **% OD (obs.)** is the headline rate: the directly measured open data rate across all articles in each funder's portfolio. *% OD (est.)* is a supplementary modeled estimate that applies journal-level head-to-head PDF vs. XML correction factors to the XML-only portion; it is informative about the magnitude of XML undercount but assumes the head-to-head subset is representative of XML-only articles within each journal. Cell shading: Total Pubs uses a blue-to-red gradient on log scale; both % OD columns share a linear blue-to-red gradient anchored on the observed range. Full rankings for all 816 funders are available in the supplementary materials on GitHub.

Funder	**Country**	**Total Pubs**	**Open Data**	**% OD (obs.)**	*% OD (est.)*
Agence Nationale de la Recherche	France	2,644	689	26.1	28.8
Horizon 2020 Framework Programme		2,408	605	25.1	28.6
National Science Foundation	USA	7,611	1,878	24.7	27.3
Wellcome Trust	UK	4,752	1,146	24.1	26.1
Swiss National Science Foundation	Switzerland	3,496	818	23.4	26.4
Deutsche Forschungsgemeinschaft	Germany	9,344	2,021	21.6	24.5
National Institutes of Health	USA	28,982	5,940	20.5	22.7
Netherlands Organisation for Scientific Research	Netherlands	1,935	381	19.7	22.9
Bundesministerium fur Bildung und Forschung	Germany	2,710	525	19.4	23.6
UK Research and Innovation	UK	14,781	2,868	19.4	23.2
Natural Sciences and Engineering Research Council of Canada	Canada	2,752	531	19.3	22.1
State Research Agency (Spain)	Spain	1,985	379	19.1	20.7
Swedish Research Council	Sweden	3,037	566	18.6	22.2
European Commission	EU	9,202	1,683	18.3	21.3
China Scholarship Council	China	1,635	273	16.7	21.2
National Health and Medical Research Council	Australia	2,406	399	16.6	20.7
Ministry of Science and Innovation (Spain)	Spain	2,555	410	16.0	18.8
European Regional Development Fund	EU	2,738	434	15.9	19.3
Canadian Institutes of Health Research	Canada	2,619	411	15.7	19.1
Japan Society for the Promotion of Science	Japan	6,680	1,009	15.1	18.5
H2020 European Research Council		2,074	300	14.5	17.6
China Postdoctoral Science Foundation	China	2,930	405	13.8	17.4
Fundacao para a Ciencia e a Tecnologia	Portugal	1,965	264	13.4	15.1
Fundacao de Amparo a Pesquisa do Estado de Sao Paulo	Brazil	1,743	232	13.3	17.7

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Funder	Country	Total Pubs	Open Data	% OD (obs.)	% OD (est.)
National Research Foundation	South Africa	3,161	411	13.0	15.8
National Key Research and Development Program	China	10,383	1,328	12.8	17.2
National Natural Science Foundation of China	China	47,356	6,018	12.7	16.8
Fundamental Research Funds for the Central Universities	China	3,155	396	12.6	17.0
National Council for Scientific and Technological Development (CNPq)	Brazil	3,479	433	12.4	17.3
National Institute for Health Research	UK	3,882	456	11.7	17.5
Natural Science Foundation of Shandong Province	China	2,031	235	11.6	14.9
Coordenacao de Aperfeicoamento de Pessoal de Nivel Superior	Brazil	3,055	341	11.2	15.9
National Research Foundation of Korea	Korea	4,854	475	9.8	13.7
Ministry of Science and ICT, South Korea	Korea	1,958	168	8.6	12.2

Table 2: Open data rates among top biomedical journals. Journals exceeding the Weibull-derived 5% survival threshold for total articles (≥1,832 articles with oddpub v7 coverage), ranked by observed open data rate. % OD (obs.) shows the directly measured rate; % OD (est.) applies journal-level correction factors from head-to-head PDF vs. XML comparison to estimate the true rate for articles with XML-only coverage. Cell shading: Total Pubs uses a blue-to-red gradient on log scale; % OD columns use a linear blue-to-red gradient. Full rankings for all 1,389 journals are available in the supplementary materials on GitHub.

Journal	Total Pubs	Open Data	% OD (obs.)	% OD (est.)
Scientific Data	2,421	1,588	65.6	65.6
iScience	3,514	1,785	50.8	50.8
Communications Biology	2,503	1,181	47.2	60.1
Nature Communications	15,129	6,835	45.2	55.9
Frontiers in Microbiology	3,850	1,443	37.5	50.2
BMC Plant Biology	1,982	668	33.7	42.5
Frontiers in Plant Science	3,649	764	20.9	30.8
Microorganisms	3,132	610	19.5	25.0
Viruses	2,076	356	17.1	26.7
Frontiers in Veterinary Science	2,238	363	16.2	23.9
Ecology and Evolution	2,325	372	16.0	55.1
Chemical Science	2,486	378	15.2	15.2
Frontiers in Immunology	5,876	859	14.6	19.8
PLoS ONE	24,182	3,427	14.2	23.8
Frontiers in Endocrinology	2,649	361	13.6	16.8
Frontiers in Nutrition	2,356	305	12.9	19.8
Scientific Reports	49,520	5,775	11.7	14.5
BMC Infectious Diseases	1,953	226	11.6	14.8
PeerJ	2,512	272	10.8	16.0

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Journal	Total Pubs	Open Data	% OD (obs.)	% OD (est.)
BMC Public Health	5,185	501	9.7	12.1
BMC Cancer	2,277	222	9.7	12.1
Frontiers in Psychology	4,285	379	8.8	12.8
Science Advances	3,594	302	8.4	16.2
Proceedings of the National Academy of Sciences	4,134	320	7.7	16.3
Frontiers in Public Health	5,387	412	7.6	14.4
Frontiers in Pharmacology	3,697	279	7.5	12.8
Heliyon	16,600	1,231	7.4	16.0
International Journal of Molecular Sciences	14,269	1,008	7.1	12.2
Animals	4,759	320	6.7	10.4
Plants	4,653	294	6.3	10.4
Frontiers in Psychiatry	2,334	142	6.1	10.2
Advanced Science	4,305	264	6.1	13.2
BMC Oral Health	2,196	112	5.1	6.6
Frontiers in Medicine	3,462	167	4.8	6.8
Frontiers in Oncology	4,069	183	4.5	6.9
RSC Advances	4,574	208	4.5	4.5
BMC Health Services Research	2,205	88	4.0	5.1
Frontiers in Neurology	2,291	86	3.8	4.8
Molecules	7,085	255	3.6	5.4
Medicine	4,702	140	3.0	16.0
ACS Omega	5,973	153	2.6	16.0
PubMed	5,728	131	2.3	15.9
Life	1,867	43	2.3	3.2
Foods	5,262	111	2.1	2.6
BMC Medical Education	2,148	46	2.1	2.7
Pharmaceuticals	1,846	35	1.9	4.2
Biomedicines	3,002	58	1.9	2.7
Cancers	4,121	77	1.9	4.3
Nutrients	4,728	92	1.9	3.7
BMJ Open	4,923	77	1.6	16.0
International Journal of Environmental Research and Public Health	2,018	18	0.9	3.1
Healthcare	3,137	21	0.7	2.5
Diagnostics	3,400	25	0.7	1.4
Medicina	2,321	11	0.5	1.9
Sensors	10,333	39	0.4	1.0
Polymers	4,477	18	0.4	1.3
Nanomaterials	2,402	7	0.3	1.2
Journal of Clinical Medicine	8,679	26	0.3	0.9
JAMA Network Open	2,805	6	0.2	15.9
Materials	8,035	13	0.2	0.6
Cureus	28,751	41	0.1	0.2
Radiology Case Reports	1,875	1	0.1	16.0
International Journal of Surgery Case Reports	2,131	1	0.0	16.0